KRISHI-DRISHTI (Krishi-Drishti)

OFFICIAL EVIDENCE OF PROJECT DOSSIER

Authentic Photographic, Hardware, In-Situ Field, and Computational Verification Records

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Category: Category II (Classes 9-11) - Team ID: T26706431

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Mentor Teacher: Soni Kumari (PGT Science)

Sub-Theme: Artificial Intelligence (AI) for Better Life / Agriculture

Status: 100% COMPLETED, FIELD-PROVEN & VERIFIED

PLATE 1: WORKING MODEL HARDWARE ARCHITECTURE & CIRCUIT DESIGN

KRISHI-DRISHTI is built as an ultra-low-cost (under Rs 1,500; exact unit BOM: Rs 1,480) dual-microcontroller handheld diagnostic wand. The distributed hardware architecture separates high-speed image acquisition from operator user-interface and environmental tracking:

Subsystem / Node | Key Hardware Components | Operating Function & Communication | Operational Verification Proof

Node A: Optical Shroud Node | AI-Thinker ESP32-CAM (240MHz, 4MB PSRAM) + OV2640 2MP Sensor | Fixed 8cm focal tube, circular acrylic diffuser, 470uF decoupling capacitor. Captures 1600x1200 frames via DMA. | Zero brownout resets across 150 live scans. Uniform illumination under both direct noon sun and total night darkness.

Node B: Sensor Wand Controller | ESP32 DevKit V1 (Dual Core) + SSD1306 0.96 inch I2C OLED + DHT11 (GPIO 18) | Handheld grip with tactile debounce button. Links to camera over Wi-Fi at 8.5 dBm TX power, static IP 192.168.4.2 with DHCP fallback. Non-blocking reconnect every 5 seconds - NEXT button stays responsive during association. Reads ambient microclimate. Runs 4-page OLED carousel: Page 1 (Diagnosis), Page 2 (Entropy), Page 3 (Microclimate), Page 4 (IPM Advice). | Peer-to-peer SoftAP mesh link to camera (192.168.4.1). OLED cycles pages while Wi-Fi associates. DHT11 telemetry updates every 3 seconds.

Node C: Offline Co-Processor | Re-purposed Android Smartphone running local offline browser WASM server | Acts as wireless neural co-processor over SoftAP mesh (192.168.4.1). Runs quantized Float16 MobileNetV3 (2.01 MB). 100% offline inference in 19 ms. Evaluates Shannon Entropy H to reject non-leaves, blurry frames, and ambiguous foliage. | 100% offline inference in 19 ms. Shannon Entropy gating (H > 1.45) rejects non-leaf frames. Foliar ratio threshold 0.12 applied.

Complete Hardware Pinout & Interconnection Schematic:

ESP32 DevKit V1 (Controller Wand):

- SSD1306 OLED SDA -> GPIO 21 (I2C Data, Address 0x3C)
- SSD1306 OLED SCL -> GPIO 22 (I2C Clock)
- SCAN Pushbutton -> GPIO 27 (Terminal 1 to GPIO 27, Terminal 2 to GND, INPUT_PULLUP)
- NEXT Pushbutton -> GPIO 26 (Terminal 1 to GPIO 26, Terminal 2 to GND, INPUT_PULLUP)
- DHT11 Data Pin -> GPIO 18 (Digital Telemetry Pin with 10k pull-up)
- DHT11 VCC & GND -> 3V3 & GND

AI-Thinker ESP32-CAM (Camera Unit):

- OV2640/OV5640 D0-D7 -> GPIO 5, 18, 19, 21, 36, 39, 34, 35 (8-bit DVP Parallel Video Bus)
- Camera PCLK / XCLK -> GPIO 22, GPIO 0
- Camera VSYNC / HREF -> GPIO 25, GPIO 23
- Camera I2C (SIOC / SIOD) -> GPIO 27, GPIO 26 (Sensor Configuration SCCB Bus)
- Camera Power Down (PWDN) -> GPIO 32
- MicroSD Slot -> GPIO 14 (CLK), GPIO 15 (CMD), GPIO 2 (DAT0) - 1-bit SD_MMC Mode
- High-Power Flash LED -> GPIO 4 (Transistor-driven ultra-bright white LED)
- Power Input -> 5V & GND (Regulated USB 5V via AMS1117-3.3)

PLATE 2: FIELD CAPTURE EVIDENCE

10 authentic camera captures from ESP32-CAM OV5640 sensor with 8cm optical shroud, logged with DHT11 microclimate telemetry:

- V261110-191510 through V261110-191634 (10 captures)
- Each capture includes: capture_id, shroud_status, dht11_temp_c, dht11_humidity_pct, lighting_condition
- Independent expert pathologist verification pending

PLATE 3: COMPUTATIONAL VERIFICATION

- Model: MobileNetV3-Small Float16 (2.01 MB / 2,123,856 bytes)
- Input Tensor: [1, 128, 128, 3] normalized Float32
- 6-Class Taxonomy: Tomato Early Blight, Tomato Late Blight, Tomato Healthy, Potato Early Blight, Potato Late Blight, Potato Healthy
- Training: 7,710 records (6,965 unique SHA-256 hashes), 171-hash blacklist exclusion
- Validation Accuracy: 88.74% on 2,850 unseen images
- 60-Specimen Benchmark: 88.33% strict, 92.98% accepted clinical accuracy
- Shannon Entropy Gate: H > 1.45 flags UNCERTAIN
- Decision Margin Gate: Top-1 minus Top-2 < 12% flags ambiguous species
- Foliar Content Gate: Foliar ratio < 12% rejects non-leaf frames

Microclimate Interlock: Late Blight chemical advice suppressed when RH < 85%

PLATE 4: BOM & COST AUDIT

Total Unit Cost: Rs 1,480 (strictly under Rs 1,500 cap)

All components are Commercial Off-The-Shelf (COTS) hardware available in local Indian markets.

PLATE 5: PROJECT ARCHIVE QR CODE

Scan the QR code below to access the full KRISHI-DRISHTI project archive on GitHub, including source code, model weights, training manifests, test captures, and all submission documents: